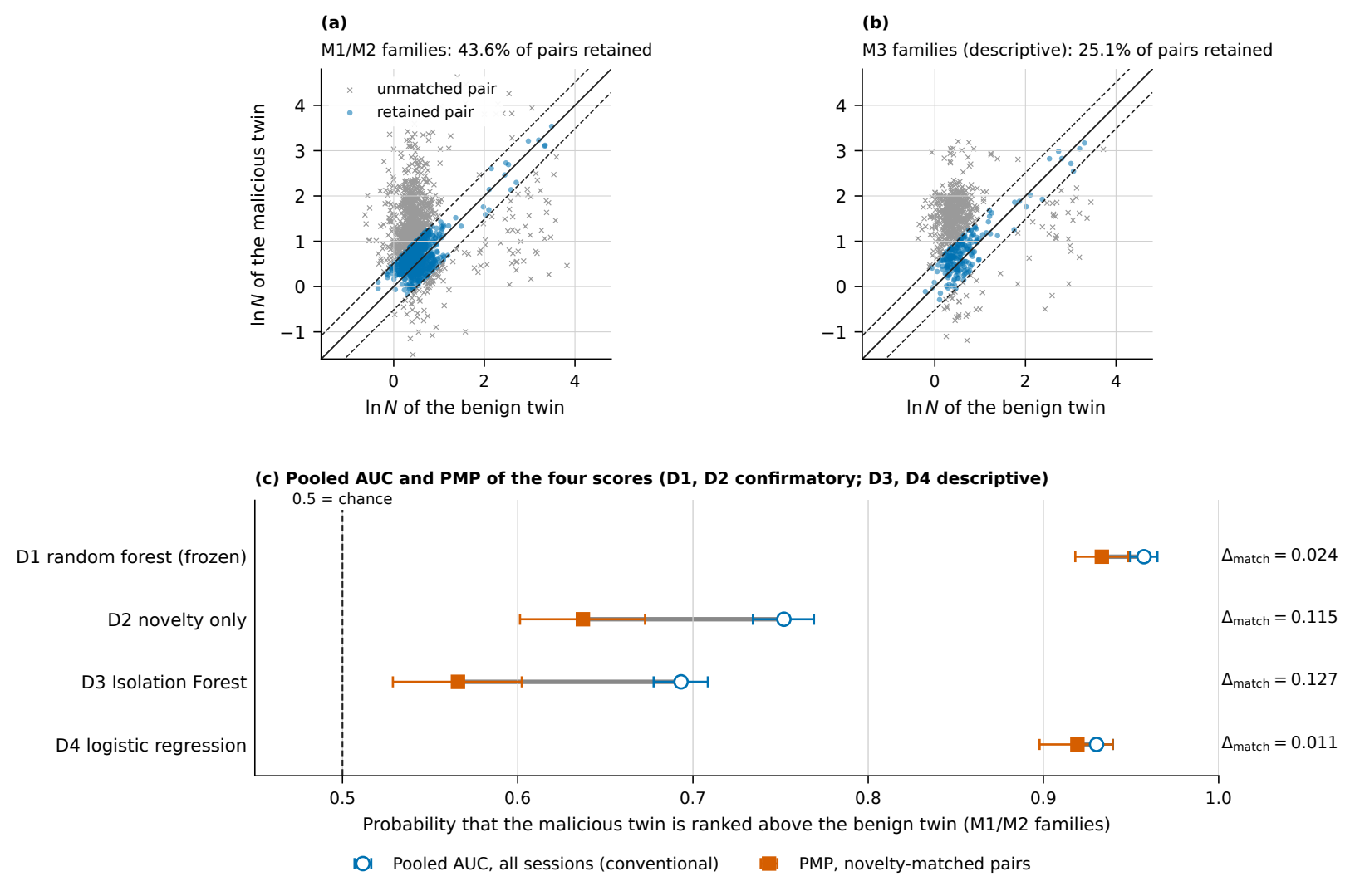

(a)
M1/M2 families: 43.6% of pairs retained
unmatched pair
retained pair
ln N of the malicious twin
ln N of the benign twin
(b)
M3 families (descriptive): 25.1% of pairs retained
ln N of the benign twin
(c) Pooled AUC and PMP of the four scores (D1, D2 confirmatory; D3, D4 descriptive)
0.5 = chance
D1 random forest (frozen)
D2 novelty only
D3 Isolation Forest
D4 logistic regression
Δmatch = 0.024
Δmatch = 0.115
Δmatch = 0.127
Δmatch = 0.011
Probability that the malicious twin is ranked above the benign twin (M1/M2 families)
Pooled AUC, all sessions (conventional)
PMP, novelty-matched pairs